

RESEARCH PROJECT (PRESENTATION)

The project must be on a topic within the theme of operating systems security. Topics must be selected by each student (first-come-first-served) to avoid substantial overlap in class projects. Suitable topics:

- Do not overlap with course topics (including discussion papers)
- Do not overlap with other student's topics (as listed in spreadsheet linked below)
- Are not too general
- Are within the scope of OS security (consider that it must related to the OS layer)

Example topics include:

1. Virtual machine introspection for host-based intrusion detection
2. A detailed description of the internal workings, algorithms used, user interface issues, etc. of a particular standalone tool for OS security enhancement. (note: you would need to specify exactly what you will do and for what standalone tool)
3. A description of the security features of an OS we have not covered in class. (note: you would need to specify which OS).

Topics can be one of the following forms:

- a. A literature review of an operating-systems related topic. In this case, the literature review must contain at least 7-10 references from reputable sources.
- b. A comprehensive review of the security features and vulnerabilities/threats of a particular operating system not covered in class.
- c. A detailed description of the internal workings, algorithms, user interface issues, etc. of a particular standalone tool for OS security enhancement.
- d. An evaluation or implementation of a software tool (or proposed software tool) that is relevant to OS security.

To claim your project topic, you must follow these instructions:

- i. Go to the "Claimed Project Topics" spreadsheet here:
<https://docs.google.com/spreadsheets/d/1b5Sjh2bwHXidcWsZLEk9f9A-4wbs7UQVKpMG1qsQ-eU/edit?gid=1670827216#gid=1670827216>
- ii. Check the list of existing project topics to ensure there is no overlap with your topic.
- iii. If you are sure your topic doesn't overlap with someone else's, you may add your name and topic to the spreadsheet.
- iv. Note: you may contact another student in the class for more clarity on their topic, in order to differentiate your project topic from theirs. It is not acceptable to modify anyone else's project topic in the spreadsheet.

Every project should contain at least the following (unless otherwise approved by the instructor for your topic), organized in the style of academic research, with the following core sections:

- Abstract or Executive Summary
- Introduction
- Literature Review (may contain various sub-sections, which are topic-dependent) or Product Review (for the cases where a specific product or set of products are being reviewed).
- Discussion and Future Work (identifies gaps in the existing research, or missing/recommended security features if the topic is a particular product).
- References (must be in IEEE format).

The presentation must be submitted as a PDF of your slide deck, where the first slide contains a link to your presentation video (an unlisted YouTube link). The presentation video must not be edited after your submission date, and must be no longer than 20 minutes in duration. Proper citations (in IEEE format) must be used for all sources. It will be graded out of 100 points as follows:

Clarity of the Introduction	10
Demonstrated understanding of the subject matter, including noting relevant relationships to course subject matter where appropriate.	30
Quality and coverage of the Literature/Product Review	25
Clarity of the Discussion and Future Work	15
Grammar and format	10
References (must be complete and use IEEE format)	10

Please refer to Canvas for the due date. Late projects will not be accepted. Medical excuses will be handled on a case-by-case basis; please contact the instructor directly regarding any medical or other reasonable excuses.